

DataLink & Product Streamer

Software Verification Report

Technical briefing | Quan Guo | Updated 22 September 2026

1. Software and Versions

Objective. Verify the software path from a synthetic data identifier (DID), through DataLink discovery, to Product Streamer delivery and independent file-integrity checks. The laboratory uses official SKA service code with simulated supporting services.

Component	Software, version and role
DataLink	Official ska-src-dm-datalink , 0.1.5 . Resolves DIDs and returns VOTable links and service descriptors. Commit: <code>997dfb03944b9c44283411264ca1c07b48b03a8e</code> .
Product Streamer	Official ska-src-dm-product-streamer-api , 0.1.2 . Delivers single files and TAR datasets. Commit: <code>27df8cb247105faf67a6d278dd9819528f211dd3</code> .
Server environment	Recorded baseline: Rocky Linux 10.2 , Python 3.13.15 , three separate Python environments, systemd process supervision.
Service dependencies	Official services: FastAPI 0.124.4 , Uvicorn 0.34.3 . Portal: FastAPI 0.141.1 , Uvicorn 0.53.0 . HTTP client: HTTPX 0.28.1 . Full dependency snapshots are included in the repository.
Laboratory integration	Custom Python portal, password gateway, process supervisor and deterministic fixture generator, with HTML/CSS/JavaScript UI. This code has no separate semantic release version; Git history identifies the published snapshot.
Python tests	Custom external client <code>test_public_e2e.py</code> : Python 3.9+ , standard library only. Additional scripts cover the internal suite, gateway and client regressions.

Test environment. <http://64.176.188.89:18080>. DataLink and Streamer run on server loopback ports 18081 and 18082 behind the laboratory gateway. The official application source is unmodified; configuration directs dependencies to local fixtures.

Simulation boundary. DMAPi, IAM, PAPI and colocated-service discovery are simulated. No real SCAPi, Rucio, StoRM or federated SRCNet service is connected. Four deterministic files are used: FITS, text, JSON and a 1 MiB binary. The dataset contains the first three files. Public HTTP is unencrypted; use synthetic data and a disposable laboratory password only.

Evaluation basis. [IVOA DataLink 1.1](#), [VOTable 1.4](#), [HTTP Semantics / RFC 9110](#) and [SHA-256 / FIPS 180-4](#). These guide selected checks; they do not establish full standards conformance.

2. Test Coverage and Reproducible Code

Test group	Checks and purpose
Access and health	Check login, origin restrictions, and DataLink/Streamer ping and health endpoints to establish protected access and service availability.
DID and address discovery	Resolve four file DIDs and one dataset DID; parse VOTables and verify advertised direct-file and Streamer URLs. Bootstrap addresses are configured, not SCAPI-discovered.
File and dataset delivery	Download four files through direct links and Streamer; compare byte counts and independently generated SHA-256 values. Inspect all three TAR members without extracting them to disk.
Partial data and errors	Verify exact 100-byte retrieval and HTTP 416 rejection; test missing-token, denied and wrong-audience synthetic identities, malformed XML, and blocked mock-token access.
Diagnostics and evidence	Inspect MIME, standardID and authentication discovery; retain address provenance, responses, downloaded fixtures and machine-readable results.

Public GitHub repository: github.com/guotsuan/datalink-streamer-lab.

The repository includes server integration code, systemd examples, dependency snapshots, Python tests and an English [README.md](#). [Deployment instructions](#) distinguish the original VM configuration from fresh setup. Run the [external test script](#) with `python3 lab/test_public_e2e.py`; enter the disposable password when prompted. Passwords, signing keys and local runtime evidence are excluded from the repository.

3. Test Results

Recorded external run: 21 September 2026, 05:11 UTC. A total of **41 HTTP requests** produced **33 checks: 30 PASS / 3 WARN / 0 FAIL**. The separate internal suite returned **14/15**, with one MIME diagnostic failure overlapping an external warning. Requests, checks and suites have different denominators.

Verified outcome. DID-to-link resolution, four-file delivery, three-member TAR integrity, exact partial bytes and the tested negative responses worked with simulated dependencies. Evidence run:

20260921T051110Z-bae7b261. The [published results summary](#) records the detailed scope and limitations.

Recorded observations. Three diagnostic warnings remain: DataLink returns `application/xml`; DataLink standardID INFO is absent; and the public unauthorized response omits WWW-Authenticate and advertises a localhost mock discovery URL. These observations are retained, not relabelled as passes.

Conclusion. The tests establish a repeatable synthetic software-verification baseline, not production readiness or full IVOA certification. Real discovery/authentication, interrupted-transfer recovery, large-file/concurrency tests and failure recovery remain unverified. The current partial-download test uses Streamer POST, not standard HTTP GET-range conformance. This report revision does not represent a new live test run.